Calibrated Multiview Inverse Finite-Element Identification of Heterogeneous Lung Mechanics with Screened Physics Auditing

Synthetic / CT-derived geometry

implanted two-level stiffness field

Calibrated multiview tracks

4 loads × 3 views × 7 frames; force and camera calibration

Learned initializer

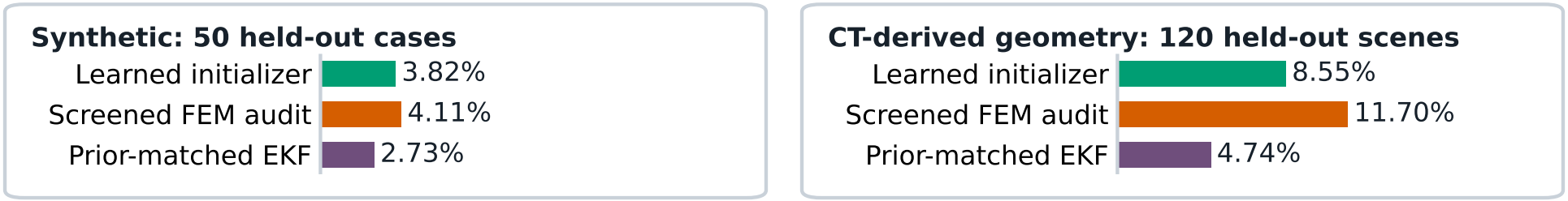
PCA–Ridge (synthetic) / hybrid MeshGNN (CT); conformal intervals

Screened FEM audit

accept if the development-fixed objective improves, else retain

Sequential EKF diagnostic

all-frame updates of E_bg and q; prior matched to the audit; no screening



Bars: median background-modulus error (%), lower is better

The accept/retain screen controls cost but does not detect model mismatch: on CT-derived geometry the audit raises the median error (8.55% → 11.70%), whereas the prior-matched sequential EKF reaches 4.74%.